

Positive Square Energy of Every Nonacyclic Cactus

26 July 2026

Abstract

Let G be a connected cactus of cyclomatic rank nine on n vertices. We prove $s^+(G) > n$. The sharp block-additive doubly nonnegative estimate reduces the problem exactly to T^8Q and T^7PP , where $T = C_3$, $P = C_5$, and Q is an arbitrary cycle. In the disconnected shared-cut case, exact colored partition audits give $66 = 63 + 3$ and $117 = 109 + 8$ proper rows. The remaining T^8Q endpoint is settled by a directly verified packing-one theorem. The $T^7P \mid P$ endpoint has $3188 = 3150 + 38$ marked classes, with sequential final-owner refinement; the two-interface $P \mid A_7 \mid P$ endpoint has $3188 = 3182 + 6$ classes and six explicit residual repairs. In the fully shared case an exact T^8Q incidence census leaves only a common-cut bouquet and, for the hostile ledger, one packing-one two-cut type. The exact T^7PP census is $8004 = 7997 + 7$; we list all seven replacements, including the leaf-pentagon opening F9, and give an executable seven-row verifier. Every separation is at an actual bridge or along proper cycle intervals, and every connector remnant, shared cut, incidence branch, and attached tree has exactly one final owner. The proof expressly uses neither the unproved two-pivot winding assertion nor the unproved candidate separator.

1 Definitions, provenance, and firewall

All graphs are finite, simple, and undirected. A cactus is a connected graph whose blocks are bridges or cycles. Its cyclomatic rank is its number of cyclic blocks. If the adjacency eigenvalues of G are $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \quad \sigma(G) = s^+(G) - |V(G)|.$$

For an induced vertex partition $V(G) = V_1 \dot{\cup} \dots \dot{\cup} V_r$, positive square energy is superadditive:

$$s^+(G) \geq \sum_i s^+(G[V_i]), \quad \sigma(G) \geq \sum_i \sigma(G[V_i]). \quad (1)$$

This is the theorem of Akbari, Kumar, Mohar, and Pragada [1]. Since a rank-nine connected graph has $|E| = n + 8$, our theorem proves, strictly, the conjectured inequality in [2, Conjecture 1.2] for this class.

Write $T = C_3$, $P = C_5$, and

$$\delta = \sec(\pi/5) - 1 = \sqrt{5} - 2 < \frac{1}{4}, \quad \delta_q = \sec(\pi/q) - 1 < 1.$$

A cycle $Q = C_q$ is *hostile* when $q \equiv 1 \pmod{4}$.

The controlling synthesis is [12], whose dependency boundary was independently subjected to the hostile audit [13]. We use the corrected octacyclic infrastructure [11], the exact rank-nine DNN

calculation [14], the disconnected marked-interface certificates [16, 18], the fully shared census [19], and the seven-row replacement verifier [21].

Two tempting statements are outside the proof. We do *not* use the two-pivot phase or winding assertion discussed in [26]: its exact two-terminal reduction does not prove the required integrated argument bound. We also do *not* use the census-free candidate separator from the rank-uniform router note [27]. Every two-interface core below is instead discharged by a finite checked router certificate or an explicit pentagon opening. No principal-arctan shortcut, hidden two-pivot estimate, edge monotonicity, connector contraction, or unproved Voronoi guard is imported.

2 Packet bounds and induced ownership

We use the established bounds, uniformly over arbitrary finite trees attached at arbitrary packet vertices,

$$\sigma(H) \geq 0 \quad \text{for connected ranks two and three,} \quad (2)$$

$$\sigma(H) > 0 \quad \text{for connected ranks four through eight,} \quad (3)$$

$$\sigma(P) \geq -\delta, \quad \sigma(Q) \geq -\delta_q \quad \text{for hostile } Q, \quad (4)$$

$$\sigma(TP) > 1 - \delta, \quad \sigma(TQ) > 0, \quad (5)$$

$$\sigma(PP) > 0, \quad \sigma(TPP) > \frac{3}{2}. \quad (6)$$

The authoritative octacyclic manuscript assembles the rank bounds and packet provenance [11, Section 2]. Their sources are the rank-two through rank-eight cactus theorems [5, 6, 7, 8, 9, 10, 11]. For one connected shared-cut cluster A_r of r triangles,

$$\sigma(A_r) > b_r, \quad (b_1, \dots, b_7) = (0, 1, 2, 3, 2, 1, 0). \quad (7)$$

The corrected matching-injection proof and recurrence, not a multiplicity-blind Sturm count, are cited in [11, Section 2].

We need two analytic packet theorems.

Lemma 2.1 (One-hostile-cycle packing one). *Suppose the cyclic blocks of a connected cactus H are one hostile $Q = C_q$ and $a \geq 1$ triangles, no two of which are vertex-disjoint. The Q may meet the triangular part at its root or be joined to that root by an arbitrary positive path. Arbitrary trees may occur everywhere, including on the path. Then*

$$s^+(H) - s^-(H) > -2\delta_q, \quad \sigma(H) > a - \delta_q.$$

Proof. Eliminate every tree off the cyclic spine using its signless matching message $M_{u \rightarrow v}(t) = t + \sum_w M_{w \rightarrow u}(t)^{-1} \geq t$. This replaces spine activities by $\alpha_v = t + y_v \geq t$ and removes a positive real factor. With $S = V(Q)$, the grouped Sachs expansion is

$$\Psi_H(t) = R + 2i(B - A),$$

where $B = Z_{H-S}(\alpha)$, $A = \sum_T Z_{H-V(T)}(\alpha)$, and

$$R = Z_H(\alpha) + 4 \sum_{T \cap Q = \emptyset} Z_{H-(S \cup V(T))}(\alpha) > 0.$$

Packing one is used exactly to exclude terms containing two triangles. If $Z_q(t)$ is the bare matching partition of Q , splitting matchings by boundary use gives $Z_H = Z_Q(\alpha|_S)B + E$, $E \geq 0$, and $Z_Q(\alpha|_S) = Z_q + L$, $L \geq 0$. Hence

$$R - Z_q(B - A) = E + LB + Z_qA + 4 \sum_{T \cap Q = \emptyset} Z_{H-(S \cup V(T))}(\alpha) > 0.$$

Thus the continuous argument of Ψ_H is strictly below the isolated- Q argument. The signed Coulson identity and $s^+(Q) - s^-(Q) = -2\delta_q$ prove the first claim. Since $|E(H)| = |V(H)| + a$ and $s^+ + s^- = 2|E|$, the second follows. This is the standalone proof of [23]; it has no all-rank decomposition premise. $\square$

Lemma 2.2 (Common-cut Schur–Sachs bounds). *For a one-vertex bouquet, uniformly over arbitrary attached trees,*

$$\sigma(T^k Q) > k - \delta_q \quad (q \equiv 1 \pmod{4}), \quad (8)$$

$$\sigma(T^k Q) > k \quad (q \text{ even or } q \equiv 3 \pmod{4}), \quad (9)$$

$$\sigma(T^k P) > k - \delta, \quad (10)$$

$$\sigma(T^k PP) > k + 1 - \frac{4}{3\sqrt{13}}. \quad (11)$$

Proof. After exact tree elimination, let A_j be the positive matching partition of the private path of lobe j and B_j the positive contribution obtained by matching the common pivot into that lobe. The normalized determinant is the scalar expression

$$W(t) = a_x + \sum_j \frac{B_j + \omega_{\ell_j}}{A_j}, \quad \omega_\ell = -2i^{-\ell}.$$

For $T^k Q$, $A_Q X > Z_q(t)$, where $X = a_x + B_Q/A_Q + \sum B_j/A_j > 0$. The positive imaginary phase, if any, is therefore strictly below that of isolated Q ; in the other congruence classes the phase is negative. Coulson integration gives (8) and (9), and $Q = P$ gives (10).

For two pentagons, adding triangles strictly increases the positive real part and decreases the imaginary part. The exact coefficient inequality

$$2R \geq t(t^4 + 7t^2 + 9)(A_1 + A_2)$$

and $\int_0^\infty 4/(t^4 + 7t^2 + 9) dt = 2\pi/(3\sqrt{13})$ give (11). The full scalar proof and the 1290-term coefficient certificate are [24, 25]. This lemma is invoked only when all named cycles really share one pivot. $\square$

Join cyclic blocks when they share a cut; the components are *shared-cut clusters*. Contracting each cluster while retaining actual bridge paths gives a tree. We repeatedly use the following ownership fact, proved in the octacyclic infrastructure [11, Lemmas 2.3–2.4].

Lemma 2.3 (Territory realization). *Connected subtrees assigned in the reduced cluster tree can be realized as connected induced territories by cutting actual bridges. If a selected cycle has two or three distinct cyclic marks, assigning the marks to nonempty proper consecutive intervals realizes connected induced territories; coincident marks remain one group. A later split refines one existing territory. Every connector remnant, shared cut, incidence branch, and off-hull tree follows its unique mark or hull attachment and has exactly one final owner.*

3 Exact DNN reduction

For an edge-containing graph set

$$\kappa(G) = \sup_{\substack{M \succ 0, M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [4, 3] says

$$\kappa(G) = b + \sum_i \kappa(C_{\ell_i}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd}, \end{cases} \quad s^-(G) \leq \kappa(G).$$

Put $\epsilon_\ell = 0$ for even ℓ and $\epsilon_\ell = \ell \tan^2(\pi/(2\ell))$ for odd ℓ . Block counting and $|E| = n + 8$ give

$$\sigma(G) \geq 8 - \sum_{i=1}^9 \epsilon_{\ell_i}. \quad (12)$$

The odd sequence decreases, $\epsilon_3 = 1$, and, for $a = \epsilon_5 = 5 - 2\sqrt{5}$,

$$3a < 2, \quad 2a > 1, \quad \epsilon_5 + \epsilon_7 < 1.$$

These comparisons have exact rational/squaring certificates in [14]. At most six triangles gives $\sum \epsilon_i \leq 6 + 3a < 8$. With seven triangles, only two pentagons can contribute at least one. With at least eight triangles the multiset is T^8Q , allowing $Q = T$. Thus the complete and exact residual list is

$$T^8Q = \{3, 3, 3, 3, 3, 3, 3, 3, q\}, \quad T^7PP = \{3, 3, 3, 3, 3, 3, 3, 5, 5\}. \quad (13)$$

Every other rank-nine cactus is strict directly from (12).

4 Disconnected shared-cut graph

The exact colored cluster-partition census [15] is

residual	all partitions	proper	direct packet rows	structural rows
T^8Q	67	66	63	3
T^7PP	118	117	109	8

Every direct row has an exact positive packet ledger. Reduced-tree pruning closes each structural row with a usable singleton triangle: a triangle leaf leaves a strict rank-eight complement; otherwise the distinguished clusters are the two leaves of a path, and the nearest singleton triangle forms a positive TQ or TP terminal packet with a strict lower-rank complement. The only endpoints left are

$$A_8 \mid Q, \quad T^7P_0 \mid P_1, \quad P_0 \mid A_7 \mid P_1. \quad (14)$$

All cuts here are actual bridge cuts, with both connector remnants assigned.

4.1 The $A_8 \mid Q$ endpoint

Mark the first cyclic-hull entry of the connector to Q . If the eight-triangle incidence is not locked at one common cut, choose an internal triangle router and include a private connector entry among its at most three marks. Split it into proper intervals. If the Q territory retains k triangles, then $k = 1$ is strict by TQ ; $k = 2$ is nonnegative with another strict triangular branch; and $3 \leq k \leq 7$ is strict by the lower-rank theorem. If $k = 0$, the other seven triangles occupy at most two branches, so one contains at least four triangles and has surplus > 3 , absorbing $\sigma(Q) > -1$.

If no such router exists, all eight triangles contain one hub. Their packing number is one even when the connector enters through a private bouquet vertex. Lemma 2.1 applies to the actual joining path and gives

$$\sigma(G) > 8 - \delta_q > 0.$$

Nonhostile Q is easier by the bridge packet ledger. This proves every disconnected T^8Q cactus and supplies the structural closure behind $66 = 63 + 3$; no two-pivot statement is involved.

4.2 Entry-locked $T^7P \mid P$: $3188 = 3150 + 38$

An internal P_0 , or an incidence-leaf P_0 entered privately, splits by proper pentagon intervals. In the remaining entry-locked family, the exact marked-entry census is

cut count	1	2	3	4	5	6	7	total
all T^7P incidences	1	9	49	145	245	205	69	723
P -leaf incidences	1	6	30	79	120	86	23	345
marked-entry classes	2	29	195	661	1144	909	248	3188
direct one-router	0	24	186	649	1134	909	248	3150
finite replacements	2	5	9	12	10	0	0	38

The 38 replacements use zero routers in two common-cut rows, one router in nine rows, two in 22, and three in five. The verifier checks every split order, proper interval size, connected retained component, common-cut or shared-cut hypothesis, both pentagonal deficits, and each private-entry cost. Its weakest ledger is

$$1 - 2\delta = 5 - 2\sqrt{5} > 0. \quad (15)$$

The corrected point is *final-owner refinement*. An interval created by an early router may provisionally contain several future packets. Every later split refines only that interval; the certificate replaces its provisional owner by the unique final packet owner. It rejects any completed row retaining a synthetic **territory:*** label. In the regression class

$$X(P())T()T()T(X(T()))T(X(T()))$$

with root cut 8, the first router's cut-8 interval, cut 8, and root all end at **packet:1**. Across all 38 replacements, every object ends at a packet owner or the explicit **naked-tree:entry** owner. Thus arbitrary trees and connector remnants follow final rather than simultaneous owners. The exact executable and complete provenance are [16, 17]. All $3188 = 3150 + 38$ classes close.

4.3 $P \mid A_7 \mid P$: $3188 = 3182 + 6$

The two labelled entries may be shared cuts, private triangle vertices, or coincident. The census has 48 unmarked seven-triangle incidence trees, 10800 ordered labelled placements before automorphisms, and 3188 canonical classes. Across all classes, the best-plan router census is exactly two

with zero routers, 3134 with one, and 52 with two. The ordinary automaton accepts 3182 of them: 3131 one-router and 51 two-router classes. Every accepted state has integer reserve at least one and hence surplus greater than (15). A private connector entry travels with its remote pentagon, so it is a pentagonal territory rather than an uncharged naked interval.

The two zero-router classes and four routed classes form the following six residual bouquet orbits, all handled by explicit repairs:

labelled interfaces	replacement	surplus
hub, hub	open one P ; packing-one T^7P arm	$> 6 - \delta$
hub, private (either order)	$P +$ common-cut T^6P	$> 6 - 2\delta$
coincident private	$A_6 + PP$	> 1
two private vertices on one T	$A_6 + P + P$	$> 1 - 2\delta$
private vertices on two T 's	$A_5 + P + P$	$> 2 - 2\delta$

The second row represents two labelled orbits, giving six in total. In the hub–hub row, open the four private vertices of one remote pentagon as a nonempty tree E , so $\sigma(E) = -1$. The complement retains seven pairwise intersecting triangles, the other pentagon, both path remnants, and all trees rooted there. Lemma 2.1 gives $\sigma(H) > 7 - \delta$, whence $\sigma(G) > 6 - \delta$. This is a one-hostile-cycle packet, not a two-pivot bound. The other rows are realized by one or two triangle intervals and the displayed packet hypotheses. Lemma 2.3 assigns every cut and branch once. The exact two-interface certificate is [18]. Therefore all $3188 = 3182 + 6$ classes close, and every disconnected T^7PP cactus is proved, completing $117 = 109 + 8$.

5 One fully shared cluster

5.1 The complete T^8Q census

The color-preserving incidence census by shared-cut count c is

Q capacity	1	2	3	4	5	6	7	8	total
$q = 3$	1	11	68	253	572	742	493	127	2267
$q = 4$	1	11	68	258	586	774	525	142	2365
$q = 5$	1	11	68	258	589	781	536	148	2392
$q = 6$	1	11	68	258	589	783	539	151	2400
$q = 7$	1	11	68	258	589	783	540	152	2402
$q \geq 8$	1	11	68	258	589	783	540	153	2403

Only eight cuts can meet Q , so the census stabilizes. Exact ordinary splits resolve all but the common-cut bouquet for the even ledger. For the hostile odd ledger they can additionally leave the unique two-cut type

$$T(X(Q))X(T())T()T()T()T()T()T()) :$$

seven triangles share a hub with a router triangle, and Q meets that router at its other cut. In both unresolved types all eight triangles contain the same hub. The bouquet satisfies Lemma 2.2; the two-cut type satisfies Lemma 2.1, including the joining segment through the router. Each gives

$$\sigma(G) > 8 - \delta_q > 0$$

for hostile Q , while nonhostile ledgers are stronger. The census and exact SAFE totals for every capacity regime are in [19, 20]. Thus every fully shared T^8Q cactus is proved.

5.2 The complete T^7PP census: $8004 = 7997 + 7$

The exact census is

	1	2	3	4	5	6	7	8	total
all	1	17	150	699	1856	2714	1998	569	8004
ordinary-split safe	0	15	148	698	1855	2714	1998	569	7997
exceptions	1	2	2	1	1	0	0	0	7

Every SAFE row has an exact legal proper-interval split and positive packet ledger. For full explicitness, the seven residual replacements and margins are

code	c	replacement	margin
N1	1	common-cut T^7PP	$> 8 - 4/(3\sqrt{13})$
F9	2	open the leaf P and retain common-cut T^7P	$> 6 - \delta$
N2	2	$P+$ common-cut T^6P	$> 6 - 2\delta$
N3	3	$P + T+$ common-cut T^5P	$> 5 - 2\delta$
N4	3	$P + P + A_5$	$> 2 - 2\delta$
N5	4	$P + P + T + A_4$	$> 3 - 2\delta$
N6	5	$P + P + T + T + A_3$	$> 2 - 2\delta$

The corresponding canonical signatures are printed separately so that each long string can break at punctuation:

N1 X(P()P()T()T()T()T()T()T())

F9 P(X(P()))X(T()T()T()T()T()T()T())

N2 T(X(P()))X(P()T()T()T()T()T()T())

N3 T(X(P()))X(P()T()T()T()T()T()T())X(T())

N4 X(T()T()T()T()T()T(X(P)))T(X(P()))

N5 X(T()T()T()T()T(X(P)))T(X(P))X(T()))

N6 X(T()T()T()T(X(P))X(T()))T(X(P))X(T())

N2–N6 use one or two legal triangle-router refinements. N1 has an actual single pivot and uses Lemma 2.2. F9 deserves a separate audit. Its seven triangles and P_0 share hub x , while leaf P_1 meets P_0 at a second cut y . Splitting router P_0 would yield only $A_7 + P_1 > -\delta$, so that split is rejected. Instead put the four private vertices of P_1 and every tree rooted there into one nonempty induced tree E , and retain y , all of P_0 , all seven triangles, and all other branches in H . Then

$$\sigma(E) = -1, \quad \sigma(H) > 7 - \delta \quad \text{by the common-cut } T^7P_0 \text{ bound,}$$

so $\sigma(G) > 6 - \delta = 8 - \sqrt{5} > 0$. The opening preserves the real pivot x ; it is not two-pivot winding.

The seven-row verifier [21, 22] regenerates all 8004 types and the exact seven exceptions. It checks every signature and labelled edge set, router mark, interval size, sequential refinement, retained packet, shared-cut hypothesis, unique cut owner, exact radical ledger, and F9's exact -1 tree charge. Hence all seven residuals close and

$$8004 = 7997 + 7$$

is a complete proof ledger, not an experimental extrapolation.

6 Main theorem and exhaustion

Theorem 6.1. *Every connected cactus G of cyclomatic rank nine, on n vertices, satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. The exact DNN inequality (12) proves every cycle multiset except (13). For a residual graph, the reduced shared-cut cluster graph either has more than one cluster or consists of one fully shared cluster. In the first case, the exact $66 = 63 + 3$ and $117 = 109 + 8$ audits reduce all rows to direct packet ledgers or the three endpoints (14). The $A_8 \mid Q$ router/packing-one proof, the $3188 = 3150 + 38$ entry-locked certificate with final-owner refinement, and the $3188 = 3182 + 6$ two-interface certificate prove those endpoints. In the second case, the complete T^8Q census and its two analytic closures prove every incidence, while the complete $8004 = 7997 + 7$ census and the seven explicit replacements prove every T^7PP incidence.

Every global separation is made at an actual bridge. Every cycle sacrifice uses nonempty proper consecutive intervals. Later splits refine one induced owner, and final-owner replacement assigns every shared cut, connector remnant, incidence branch, and attached tree exactly once. The two pentagon openings record their exact nonempty-tree cost -1 . Common-cut estimates are used only at a real common pivot, and packing one only after all relevant triangles are directly seen to contain one hub. No unproved two-pivot winding statement and no candidate separator occurs. Thus every case has $\sigma(G) > 0$. $\square$

A Exact reproduction

The certificate programs require Python 3.10 or newer. The first five use the standard library and exact integer or `Fraction` arithmetic; the final common-cut coefficient check requires SymPy 1.14.0. From the repository root, create a self-contained local environment and run

```
python3 -c 'import sys; sys.exit("Python 3.10+ required") \
    if sys.version_info < (3, 10) else None'
python3 -m venv .venv-nonacyclic
.venv-nonacyclic/bin/python -m pip install sympy==1.14.0
.venv-nonacyclic/bin/python research/rank-nine-cactus-residual-census.py
.venv-nonacyclic/bin/python \
    research/nonacyclic-fully-shared-incidence-census.py
.venv-nonacyclic/bin/python \
    research/nonacyclic-t7p-last-bridge-conservative.py
.venv-nonacyclic/bin/python \
    research/nonacyclic-t7-two-interface-census.py
.venv-nonacyclic/bin/python \
    research/nonacyclic-t7pp-seven-exceptions-resolution.py
.venv-nonacyclic/bin/python \
    positive-square-energy/experiments/c5_bouquet_matching_certificate.py
```

The commands respectively reproduce the DNN residuals and $67/66/63/3$, $118/117/109/8$ partition counts; the fully shared T^8Q and $8004 = 7997 + 7$ censuses; $3188 = 3150 + 38$ with final owners; the independent $3188 = 3182 + 6$ two-interface census; all seven replacement rows including F9; and the frozen 1290-term common-cut coefficient certificate. The hardened $T^7P \mid P$, two-interface, seven-exception, and coefficient executables use explicit checks and fail closed even under `python`

-O. The residual partition and fully shared census generators still use Python `assert`; run those two exactly as shown, without -O. The programs certify finite enumeration, routing, ownership data, and exact ledgers; the analytic packet inequalities are the mathematical lemmas cited above.
Build with

```
bash scripts/build-paper.sh all-nonacyclic-cacti
```

or directly with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \  
-cd all-nonacyclic-cacti/paper.tex
```

References

- [1] S. Akbari, H. Kumar, B. Mohar, and S. Pragada, *A linear lower bound for the square energy of graphs*, Electron. J. Combin. 32(3) (2025), Paper P3.53.
- [2] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.
- [3] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.
- [4] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, 2026; repository path `sharp-cactus-dnn/paper.tex`.
- [5] Companion manuscript, *Positive Square Energy of Every Bicyclic Cactus*, 2026; `all-bicyclic-cacti/paper.tex`.
- [6] Companion manuscript, *Positive Square Energy of Every Tricyclic Cactus*, 2026; `all-tricyclic-cacti/paper.tex`.
- [7] Companion manuscript, *Positive Square Energy of Every Tetracyclic Cactus*, repaired theorem, 2026; `all-tetracyclic-cacti/paper.tex`.
- [8] Companion manuscript, *Positive Square Energy of Every Pentacyclic Cactus*, 2026; `all-pentacyclic-cacti/paper.tex`.
- [9] Companion manuscript, *Positive Square Energy of Every Hexacyclic Cactus*, 2026; `all-hexacyclic-cacti/paper.tex`.
- [10] Companion manuscript, *Positive Square Energy of Every Heptacyclic Cactus*, 2026; `all-heptacyclic-cacti/paper.tex`.
- [11] Companion manuscript, *Positive Square Energy of Every Octacyclic Cactus*, authoritative corrected infrastructure, 26 July 2026; `all-octacyclic-cacti/paper.tex`.
- [12] *Positive square energy of every connected rank-nine cactus*, corrected synthesis, 26 July 2026; `research/nonacyclic-cactus-complete-synthesis-2026-07-26.md`.
- [13] *Hostile audit verdict: connected rank-nine cacti*, 26 July 2026; `research/rank-nine-cacti-hostile-audit-verdict-2026-07-26.md`.

- [14] *Exact sharp-DNN residual cycle multisets for rank-9 cacti*, 26 July 2026; `research/nonacyclic-dnn-residuals-2026-07-26.md`.
- [15] Exact executable, `research/rank-nine-cactus-residual-census.py`.
- [16] *Disconnected rank-nine T^7PP : exact marked-interface reduction*, 26 July 2026; `research/nonacyclic-disconnected-t7pp-marked-interface-status-2026-07-26.md`.
- [17] Exact executable, `research/nonacyclic-t7p-last-bridge-conservative.py`.
- [18] Exact two-interface certificate and note, `research/nonacyclic-t7-two-interface-census.py` and `research/nonacyclic-t7-two-interface-census-2026-07-26.md`.
- [19] *Rank-nine fully shared incidence census: T^8Q and T^7PP* , 26 July 2026; `research/nonacyclic-fully-shared-incidence-census-2026-07-26.md`.
- [20] Exact executable, `research/nonacyclic-fully-shared-incidence-census.py`.
- [21] *The seven fully shared nonacyclic T^7PP exceptions*, 26 July 2026; `research/nonacyclic-t7pp-seven-exceptions-resolution-2026-07-26.md`.
- [22] Exact seven-row verifier, `research/nonacyclic-t7pp-seven-exceptions-resolution.py`.
- [23] *Rooted packing-one hostile-cycle lemma for octacyclic provenance*, 26 July 2026; `research/octacyclic-packing-one-hostile-cycle-lemma-2026-07-26.md`.
- [24] *Common-cut residual bouquets: an exact rooted Schur–Sachs inequality*, 26 July 2026; `research/common-cut-bouquet-rooted-schur-2026-07-26.md`.
- [25] Exact coefficient verifier, `positive-square-energy/experiments/c5_bouquet_matching_certificate.py`.
- [26] *Two-pivot Schur–Sachs triangular cactus reduction, open winding boundary*, 26 July 2026; `research/two-pivot-schur-sachs-triangular-cactus-2026-07-26.md`.
- [27] *Rank-uniform triangular router interface theorem*, `candi-date separator explicitly excluded`, 26 July 2026; `research/rank-uniform-triangular-router-interface-theorem-2026-07-26.md`.